

This assignment will not be graded and is only for practice.

Level 1

1) Let $f(x, y, z) = (xy, x + y + z^2)$, where $x, y, z \in \mathbb{R}$. Choose the set of correct options.

1 point

- ☐ The domain of f is $\mathbb{R}^2$ and the co-domain of f is $\mathbb{R}^2$
- ☐ The domain of f is $\mathbb{R}^3$ and the co-domain of f is $\mathbb{R}^2$
- ☐ f is continuous everywhere in its domain
- ☐ f is not continuous anywhere in its domain

No, the answer is incorrect.

Score: 0

Accepted Answers:

The domain of f is $\mathbb{R}^3$ and the co-domain of f is $\mathbb{R}^2$
 f is continuous everywhere in its domain

2) Let $f(x, y) = \begin{cases} \frac{\cos y \sin x}{x} & x \neq 0 \\ \cos y & x = 0 \end{cases}$.

1 point

Choose the correct options.

- ☐ f is not defined at $(0, 0)$
- ☐ f is defined at $(0, 0)$
- ☐ f is continuous at $(0, 0)$
- ☐ f is not continuous at $(0, 0)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

f is defined at $(0, 0)$
 f is continuous at $(0, 0)$

3) Let $h(u, v) = \sin(uv)$, $g(x) = x^3$, $f(y) = \cos y$. Let $p(x, y) = h(g(x), f(y))$

1 point

Choose the correct option(s).

- ☐ p is a scalar-valued function
- ☐ p is a vector-valued function
- ☐ p is continuous everywhere since $g.f$ is continuous everywhere in its domain
- ☐ p is continuous everywhere, g is also continuous everywhere while f is not continuous everywhere in its domain

No, the answer is incorrect.

Score: 0

Accepted Answers:

p is a scalar-valued function

p is continuous everywhere since $g.f$ is continuous everywhere in its domain

4) Let $p(t) = (\ln(1 - t), \frac{1}{t}, 3t)$.

1 point

Choose the incorrect option(s).

- ☐ p is continuous for all real t less than 1 but not equal to zero
- ☐ p is continuous on $\mathbb{R}$
- ☐ p is continuous in $\{t \in \mathbb{R} \mid t < 1\}$
- ☐ p is continuous in $\{t \in \mathbb{R} \mid t = 1, t = 0\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

p is continuous on $\mathbb{R}$

p is continuous in $\{t \in \mathbb{R} \mid t < 1\}$

p is continuous in $\{t \in \mathbb{R} \mid t = 1, t = 0\}$

5) Let $f(x, y) = \ln(x^2 + y^2 - 1)$. Identify where f is continuous. Note: Equation of a circle with radius r and center (a, b) is given by $(x - a)^2 + (y - b)^2 = r^2$ **1 point**

- ☐ Everywhere in $\mathbb{R}^2$
- ☐ Everywhere in $\mathbb{R}^2 \setminus \{(1, 0), (0, 1)\}$
- ☐ Everywhere in $\mathbb{R}^2$ outside the circle with radius 1 and center $(0, 0)$
- ☐ None of the above

No, the answer is incorrect.

Score: 0

Accepted Answers:

Everywhere in $\mathbb{R}^2$ outside the circle with radius 1 and center $(0, 0)$

Level 2

6) Let $g(u, v, w) = \frac{uv}{w}$ and $f(x, y, z) = g(P(x, y, z), Q(x, y, z), R(x, y, z))$ **1 point**

where $P(x, y, z) = e^{x^2+y}$, $Q(x, y, z) = \sqrt{y^2 + z^2 + 3}$, $R(x, y, z) = \sin(xyz) + 5$.

Choose the correct option(s).

- ☐ f is a vector-valued function
- ☐ f is a scalar-valued function
- ☐ f is continuous everywhere in its domain.
- ☐ P is continuous everywhere in its domain, while Q is not continuous everywhere in its domain

No, the answer is incorrect.

Score: 0

Accepted Answers:

f is a scalar-valued function

f is continuous everywhere in its domain.

7) Let

1 point

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}, f(x, y) = \begin{cases} \frac{x^2}{x^2+y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}.$$

Choose the correct option(s).

- ☐ $\lim_{(x,y) \rightarrow (0,0)} f$ does not exist
- ☐ $\lim_{(x,y) \rightarrow (0,0)} f$ exists
- ☐ f is continuous at $(0, 0)$
- ☐ f is not continuous at $(0, 0)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\lim_{(x,y) \rightarrow (0,0)} f$ does not exist
 f is not continuous at $(0, 0)$

8) Let

1 point

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}, f(x, y) = \begin{cases} \frac{xy}{x^2+y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}.$$

Choose the correct option(s).

- ☐ $\lim_{(x,y) \rightarrow (0,0)} f$ does not exist
- ☐ $\lim_{(x,y) \rightarrow (0,0)} f$ exists
- ☐ f is continuous at $(0, 0)$
- ☐ f is not continuous at $(0, 0)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\lim_{(x,y) \rightarrow (0,0)} f$ does not exist

f is not continuous at $(0, 0)$

9) Let $f(r) = \frac{1}{\langle r, r \rangle - 1}$, where $r = (x, y, z)$ and the inner product $\langle, \rangle$ is the dot product **1 point**
Choose the set of incorrect options.

- ☐ f is continuous throughout $\mathbb{R}^3$
- ☐ f is continuous throughout $\mathbb{R}^3 \setminus \{(0, 0, 0)\}$
- ☐ f is discontinuous on the unit sphere $\{r \in \mathbb{R}^3 : \|r\| = 1\}$
- ☐ f is continuous throughout $\mathbb{R}^3 \setminus \{(1, 0, 0)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

f is continuous throughout $\mathbb{R}^3$
 f is continuous throughout $\mathbb{R}^3 \setminus \{(0, 0, 0)\}$
 f is continuous throughout $\mathbb{R}^3 \setminus \{(1, 0, 0)\}$

10) Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}, f(x, y) = \begin{cases} \frac{x^k y}{x^4 + y^4} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$. Find the minimum possible positive integer value of k such that f is continuous at $(0, 0)$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 4

1 point

Your score is: 0/10